

Quantum Gravitation

Lecture 4

Massive Spin-Two Representations

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$h_{\mu\nu} = h_{\nu\mu}, h^\mu_\mu = 0$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$k^\mu h_{\mu\nu} = 0$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$\sum_{s=1}^5 e_{\mu\nu}^s e_{\rho\sigma}^{s*} = \Pi_{\mu\nu,\rho\sigma}$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Seo, ch. 4; Spin-Two Polarization Tables; Q-Registry Q-11–Q-18